MEET PATEL

Availability July 2026 | Jersey City (Willing to relocate) | +1 (201) 359-1312 | Portfolio | meetpatel.aiml@gmail.com | LinkedIn

EDUCATION

Master's, Computer Science at **Fairleigh Dickinson University**, New Jersey — May 2026
Master's, Computer Engineering at **Ganpat University**, India — December 2024

SKILLS

- **Languages:** Python, SQL, JavaScript / TypeScript
- **ML Frameworks & Inference:** PyTorch, TensorFlow, ONNX Runtime, CoreML, TensorRT, vLLM
- **Model Optimization:** INT8 Post-Training Quantization, Execution Graph Compilation, Hardware-Accelerated Inference (TensorRT)
- **Architectures & Systems:** VLMs & Multi-Modal Systems, Distributed ML Architectures, Context-Aware AI Systems
- **MLOps & Cloud:** Google Cloud Platform (GCP), Docker, Kubernetes, FastAPI, Jenkins

EXPERIENCE

Deep Learning Intern — May 2026 to Present
New Jersey Department of Health (DOH)

- Independently architected an end-to-end Computer Vision QC pipeline for blood specimen images, saving 2.5 hours of manual review daily across ∼150 dried blood spot specimens.
- Benchmarked and combined OpenCV and YOLOv8m for specimen segmentation, balancing minimal memory requirements with high accuracy to overcome unstructured scanned artifacts, improving extraction from 92% to 99.5%.
- Implemented a hybrid architecture with ResNet50_2 and PatchCore for anomaly detection. Achieved 98% accuracy and 0.90 F1, utilizing PatchCore anomaly maps for biological explainability. Enforced fully local inference to meet HIPAA data sensitivity requirements.

Research Assistant — January 2025 to May 2026
Fairleigh Dickinson University, New Jersey

- Achieved the highest model performance in a competitive multi-author benchmark using fine-tuned VGG16 on 10,000+ dermoscopic images (7-class, HAM10000). Reached 97.75% validation accuracy and F1 = 0.9547, outperforming baseline EfficientNetV2 (91.32%) and ViT-B/16 (94.2%).
- Resolved severe class imbalance via inverse-frequency weighted CrossEntropyLoss and stabilized test fluctuations using ReduceLROnPlateau scheduling and gradient clipping. Reduced train-val gap to 0.20%.

Machine Learning Intern — January 2023 to April 2023
Tri State Technologies, Ahmedabad, India

- Improved production model accuracy by 17% after optimizing the feature extraction pipeline. Resolved feature conflicts and upstream noise contamination that degraded model performance.
- Managed an end-to-end ML stack via multi-endpoint FastAPI service deployed on GCP and Modal.com for scalable, serverless GPU execution and Jenkins CI/CD.

PROJECTS

Privacy Preserving Yoga Recommendation Model — PyTorch, CoreML, Two-Tower Architecture – Code | Demo

- Built a 4-stage LLM data extraction pipeline to collect data of 1,345 yoga poses from 50+ raw PDFs, generating 282,207 training samples via a rule-based system, including medical constraint layers to ensure training data was grounded in verified literature.
- Architected a 100% on-device personalized yoga prediction application, purely using watch-sensor data via two-tower approach, executing inference locally. Pre-computed static yoga embeddings and generated live user embeddings for <20ms cosine similarity search.
- Deployed the system asymmetrically to the ANE, orchestrating heavy pre-computation for the yoga tower (384-dim to 32-dim) and lightweight live inference for watch sensor data (17-dim to 32-dim), achieving a Recall@10 of 0.959 and NDCG@10 of 0.809 on cold-start validation.

Real-Time Multimodal Q&A System — PyTorch, VLM, ONNX Runtime – Code | Demo

- Architected a low-latency live video question answering system from handwritten questions using FastViTHD and Qwen2, compiled into an ONNX execution graph. Executed INT8 Post-Training Quantization and achieved a 4x latency reduction (sub 500ms TTFT) with a <1.2% degradation in zero-shot accuracy.
- Engineered a decoupled architecture to independently scale the FastViTHD encoder and language model. Implemented aggressive Lanczos downscaling algorithm to restrict inputs to a single processing patch, reducing vision-tower compute by 80%.

Neuromorphic vs. Transformer NLP Benchmarking — PyTorch, snnTorch, Hardware Optimization – Code | Demo

- Implemented an SNN compression pipeline based on NVIDIA's 2:4 structured sparsity, mapping dense linear layers to PyTorch Semi-Structured tensors which can engage A100 Tensor Cores and reduce the memory footprint.
- Architected a low-power NLP microservice, replacing $O(N^2)$ Transformer attention with binary additions, achieving a ∼150x reduction in compute energy with 89% accuracy retention.